

1. Schema Loaded

Description: Verified that all tables were created successfully in the momo_test database.

```
ahmad@Pavilion14: ~/Momo x + v
mysql> CREATE DATABASE momo_test;
Query OK, 1 row affected (0.26 sec)

mysql> USE momo_test;
Database changed
mysql> SOURCE database/database_setup.sql;
Query OK, 0 rows affected (0.29 sec)

Query OK, 0 rows affected (0.08 sec)

Query OK, 0 rows affected (0.09 sec)

Query OK, 0 rows affected (0.05 sec)

Query OK, 0 rows affected (0.08 sec)

Query OK, 0 rows affected (0.16 sec)

Query OK, 0 rows affected (0.11 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_momo_test |
+-----+
| CategoryGroups      |
| Services            |
| SystemLogs          |
| TransactionCategories|
| TransactionUsers    |
| Transactions         |
| Users               |
+-----+
7 rows in set (0.07 sec)
```

```
ahmad@Pavilion14: ~/Momo x + v
mysql> DESCRIBE Transactions;
+-----+-----+-----+-----+-----+-----+
| Field                | Type                | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| transactionId        | char(36)             | NO   | PRI | NULL    |       |
| externalTransactionId | varchar(50)          | YES  | UNI | NULL    |       |
| amount               | decimal(10,2)        | YES  |     | NULL    |       |
| currency              | varchar(5)           | YES  |     | NULL    |       |
| transactionDate       | datetime             | YES  |     | NULL    |       |
| transactionType       | enum('INCOMING','OUTGOING') | YES  |     | NULL    |       |
| fee                   | decimal(10,2)        | YES  |     | NULL    |       |
| balanceAfter          | decimal(10,2)        | YES  |     | NULL    |       |
| categoryId            | char(36)             | YES  | MUL | NULL    |       |
| serviceId             | char(36)             | YES  | MUL | NULL    |       |
| logId                | char(36)             | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
11 rows in set (0.11 sec)

mysql> DESCRIBE TransactionUsers;
+-----+-----+-----+-----+-----+-----+
| Field                | Type                | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| transactionId        | char(36)             | NO   | PRI | NULL    |       |
| userId               | char(36)             | NO   | PRI | NULL    |       |
| role                 | enum('SENDER','RECEIVER') | NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)

mysql>
```

2. Test Data Inserted

Description: Inserted test data into all tables.

```
ahmad@Pavilion14: ~/Momo x
+ + v
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| momo_test  |
+-----+
1 row in set (0.00 sec)

mysql> SOURCE database/testing/test_data.sql;
Query OK, 5 rows affected (0.06 sec)
Records: 5 Duplicates: 0 Warnings: 0

Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> |
```

3. Full Transaction JOIN

Description: Displayed transactions with sender, receiver, category, group, and service information. Confirms that foreign key relationships are working.

```
ahmad@Pavilion14: ~/Momo x + -
```

```
mysql> SELECT
->   t.transactionId,
->   u1.fullName AS sender,
->   u2.fullName AS receiver,
->   tc.categoryName,
->   cg.groupName,
->   s.serviceName,
->   t.amount,
->   t.currency,
->   t.balanceAfter
-> FROM Transactions t
-> JOIN TransactionUsers tu1 ON t.transactionId=tu1.transactionId AND tu1.role='SENDER'
-> JOIN Users u1 ON tu1.userId=u1.userId
-> JOIN TransactionUsers tu2 ON t.transactionId=tu2.transactionId AND tu2.role='RECEIVER'
-> JOIN Users u2 ON tu2.userId=u2.userId
-> JOIN TransactionCategories tc ON t.categoryId=tc.categoryId
-> JOIN CategoryGroups cg ON tc.groupId=cg.groupId
-> JOIN Services s ON t.serviceId=s.serviceId;
```

transactionId	sender	receiver	categoryName	groupName	serviceName	amount	currency	balanceAfter
t1	Alice Mukamana	Jean Habimana	P2P Transfer	TRANSFER	MoMo Transfer	5000.00	RWF	45000.00
t2	Alice Mukamana	MTN MoMo System	Merchant Payment	PAYMENT	Local Merchant	10000.00	RWF	34900.00
t3	Alice Mukamana	MTN MoMo System	Airtime Purchase	AIRTIME	MTN Airtime	1500.00	RWF	33400.00
t4	Eric Niyonzima	MTN MoMo System	Electricity	UTILITY	REG Utility	20000.00	RWF	10170.00
t5	Claire Uwimana	Alice Mukamana	Data Bundle	DATA	MTN Data	3000.00	RWF	30370.00

```
5 rows in set (0.08 sec)

mysql> |
```

4. Transactions Per User

Description: Total transactions per user. Demonstrates many-to-many relationships.

```
mysql> SELECT u.fullName, COUNT(tu.transactionId) AS totalTransactions
FROM Users u
JOIN TransactionUsers tu ON u.userI    -> FROM Users u
-> JOIN TransactionUsers tu ON u.userId=tu.userId
-> GROUP BY u.fullName;
+-----+
| fullName | totalTransactions |
+-----+
| Alice Mukamana | 4 |
| Jean Habimana | 1 |
| Eric Niyonzima | 1 |
| MTN MoMo System | 3 |
| Claire Uwimana | 1 |
+-----+
5 rows in set (0.02 sec)

mysql> |
```

5. Update Query

Description: Updated the fee for transaction t1 and confirmed the update.

```
mysql> UPDATE Transactions SET fee=75 WHERE transactionId='t1';
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT transactionId, fee FROM Transactions WHERE transactionId='t1';
+-----+
| transactionId | fee |
+-----+
| t1            | 75.00 |
+-----+
1 row in set (0.01 sec)

mysql> |
```

6. Foreign Key Constraint Test

Description: Attempted to delete a SystemLogs row referenced by Transactions. Error proves reference integrity works.

```
mysql> DELETE FROM SystemLogs WHERE logId='l5';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails ('momo_test`.`Transactions`, CONSTRAINT `Transactions_ibfk_3` FOREIGN KEY (`logId`) REFERENCES `SystemLogs` (`logId`))
mysql> DELETE FROM SystemLogs WHERE logId='l5';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails ('momo_test`.`Transactions`, CONSTRAINT `Transactions_ibfk_3` FOREIGN KEY (`logId`) REFERENCES `SystemLogs` (`logId`))
mysql> |
```

7. ENUM Constraint Test

Description: Attempted to insert a user with an invalid ENUM value. Error proves ENUM constraints are enforced.

```
mysql> INSERT INTO Transactions VALUES
-> ('tx', 'EXT999', 1000, 'RWF', NOW(), 'OUTGOING', 10, 900, 'BAD', 's1', 'l1');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('momo_test`.`Transactions`, CONSTRAINT `Transactions_ibfk_1` FOREIGN KEY (`categoryId`) REFERENCES `TransactionCategories` (`categoryId`))
mysql> INSERT INTO Users VALUES
-> ('u99', 'Invalid', '0700000000', 'ADMIN');
ERROR 1265 (01000): Data truncated for column 'userType' at row 1
mysql> |
```